

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Article Review

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO 4: Students will be able to analyze and evaluate advanced deep learning techniques and architectures, interpret research findings, and apply appropriate models to real-world problems. They will be able to critically review research articles, compare methodologies, and justify suitable deep learning solutions. (BL-6)

Assessment Tool Weightage: 30%

QUESTIONS:

Q1. Article Identification and Summary

Select a recent peer-reviewed research article related to Deep Learning, such as CNNs, Transfer Learning, ResNet, DenseNet, RNNs, LSTM, Autoencoders, or other advanced architectures. Provide the article title, authors, publication venue, year, and research objective. Summarize the main problem addressed, proposed approach, and key findings in your own words.

Q2. Problem Statement and Motivation

Explain the real-world problem addressed by the selected article and analyze why the problem requires a deep learning solution. Identify the limitations of existing approaches discussed by the authors and explain the motivation for the proposed method.

Q3. Methodology and Architecture Review

Critically review the methodology used in the article. Describe the dataset, preprocessing, model architecture, important layers or components, training strategy, and evaluation metrics. Illustrate the proposed architecture or workflow and explain the role of its major components.

Q4. Results, Comparison and Critical Analysis

Analyze the experimental results reported in the article. Compare the proposed method with the baseline or existing methods using the reported performance measures. Discuss the strengths, limitations, computational considerations, and validity of the results. Identify at least two possible improvements or future research directions.

Q5. Application and Personal Evaluation

Explain how the reviewed deep learning method can be applied to a practical industry problem. Propose a suitable implementation or extension and justify your choices. Conclude the review with your overall evaluation of the article, including its contribution, practical relevance, and scope for future work.

ARTICLE REVIEW

Deep Learning-Based Steel Surface Defect Detection Using YOLOv5-KBS

Title: Research on steel surface defect classification method based on deep learning

Authors: Yang Gao, Gang Lv, Dong Xiao, Xize Han, Tao Sun, and Zhenni Li

Journal: Scientific Reports, Volume 14, Article 8254

Publisher: Springer Nature

Year: 2024

DOI: 10.1038/s41598-024-58643-1

The article was published in *Scientific Reports* on 8 April 2024. It proposes an improved YOLOv5-based model called YOLOv5-KBS for detecting surface defects in steel. The proposed model combines Squeeze-and-Excitation (SE) attention, Bidirectional Feature Pyramid Network (BiFPN), and K-means-based anchor-box optimization. The authors report a 4.2% improvement in mAP compared with the YOLOv5 baseline and a detection speed of approximately 70 FPS.

Abstract

Steel surface defect detection is an important quality-control task in modern manufacturing industries. Traditional manual inspection methods are time-consuming, labor-intensive, and may produce inconsistent results. This article reviews a deep learning-based steel surface defect detection method that improves the YOLOv5 object detection framework through K-means anchor-box optimization, Squeeze-and-Excitation (SE) attention, and Bidirectional Feature Pyramid Network (BiFPN). The proposed YOLOv5-KBS model is designed to detect different types of steel surface defects, including scratches, inclusions, crazing, patches, pitted surfaces, and rolled-in scale.

The model is evaluated using the NEU-DET dataset, which contains 1,800 images across six defect categories. Experimental results show that the proposed approach improves detection performance compared with the baseline YOLOv5 model while maintaining approximately real-time processing speed. The combination of attention mechanisms and multi-scale feature fusion helps the model identify defects

of different sizes and improves its ability to focus on relevant surface features. The review also analyzes the strengths, limitations, computational considerations, and potential industrial applications of the method. Overall, the study demonstrates that deep learning-based object detection can provide an effective foundation for automated steel quality inspection and has significant potential for Industry 4.0 manufacturing systems.

Q1. Article Identification and Summary

1.1 Research Objective

The main objective of the article is to develop a fast and accurate deep learning system for automatically detecting surface defects in steel products. Steel manufacturing requires continuous inspection because defects can reduce product quality and may lead to financial losses or safety problems.

The authors selected the YOLO family of object detection models because YOLO can perform detection and localization in a single-stage process, making it suitable for real-time industrial applications. The researchers first compared YOLOv5, YOLOX, and YOLOv7 and found that YOLOv5 provided a useful balance between detection accuracy and speed for the selected steel-defect dataset.

The researchers then developed **YOLOv5-KBS**, an improved version of YOLOv5. The major modifications were:

1. Addition of an SE attention mechanism.
2. Replacement of PANet with BiFPN.
3. Generation of more suitable anchor boxes using K-means clustering.
4. Use of data augmentation techniques to improve generalization.

The final model was designed to recognize steel defects of different sizes while reducing the effect of complex backgrounds.

1.2 Main Problem Addressed

The research addresses the problem of automatic inspection of steel surfaces. Traditional inspection can involve human visual inspection and conventional image-processing techniques. These approaches can be affected by worker fatigue, environmental conditions, noise, dust, and variations in defect appearance.

Steel defects can also differ significantly in:

- Shape
- Size
- Texture
- Location

- Appearance
- Background conditions

Therefore, a model that works well for one defect type may not perform equally well for another. Deep learning provides an advantage because it can learn useful visual features directly from image data rather than requiring engineers to manually design every feature.

1.3 Main Findings

The article reports that the YOLOv5 baseline achieved an average mAP of 75.7% on the selected experimental setup, while the improved YOLOv5-KBS model achieved a 4.2 percentage-point improvement in mAP over the baseline. The proposed model also achieved approximately 70 FPS, making it suitable for real-time or near-real-time industrial inspection.

The authors also report particularly noticeable improvements for some defect categories. Detection accuracy increased by 10.8% for scratches and 9.6% for pitted-surface defects, while crazing and patches also showed improvements.

Q2. Problem Statement and Motivation

2.1 Real-World Problem

Steel sheets and strips are widely used in construction, automobiles, machinery, and other industrial applications. During manufacturing, defects can appear on the steel surface. Examples include:

- Crazing
- Inclusion
- Patches
- Pitted surface
- Rolled-in scale
- Scratches

The presence of these defects can affect product quality. Therefore, manufacturers need reliable inspection systems that can identify defective regions quickly.

The paper explains that conventional inspection approaches depend significantly on human expertise and can require considerable time and manpower. Industrial environments may also contain dust, noise and other disturbances, which can make manual inspection difficult.

2.2 Why Deep Learning is Required

A deep learning approach is suitable because steel defects can have complicated visual characteristics. A simple thresholding or edge-detection algorithm may work for a specific defect but fail when lighting, texture or defect shape changes.

CNN-based deep learning models can automatically learn hierarchical features.

For example:

Low-level features → edges and textures

Middle-level features → defect patterns

High-level features → complete defect representations

This allows the system to identify defects without requiring extensive manual feature engineering.

2.3 Limitations of Existing Approaches

Traditional computer vision methods generally require manually designed features. Different types of defects may require different feature extraction techniques. This limits portability and adaptability.

The article also discusses existing deep learning methods and notes that increasing detection accuracy can sometimes result in reduced detection speed. Industrial inspection, however, requires both:

High accuracy + High processing speed

Another difficulty is that steel defects can have substantially different sizes. Small defects may be missed, while large defects can be confused with background patterns.

Therefore, the researchers were motivated to improve both feature extraction and multi-scale feature fusion while maintaining real-time performance.

Q3. Methodology and Architecture Review

3.1 Dataset

The researchers used the NEU-DET dataset for their experiments.

The dataset contains:

Parameter	Details
Dataset	NEU-DET
Total images	1,800
Image size	200 × 200 pixels
Number of classes	6
Images per class	300

Parameter	Details
Training/testing split	9:1
Annotation	XML converted to YOLO-compatible text format

The six defect categories are:

1. Crazing
2. Inclusion
3. Patches
4. Pitted surface
5. Rolled-in scale
6. Scratches

The dataset was collected using a line-array camera in a factory environment.

3.2 Preprocessing

The YOLOv5 input process includes image resizing and preprocessing. The paper describes **letterboxing**, which resizes an image while maintaining its aspect ratio.

The authors also use data augmentation techniques such as:

- Cropping
- Flipping
- Rotation
- Deformation
- Splicing
- MixUp

These methods increase variation in the training data and help reduce overfitting.

3.3 YOLOv5 Architecture

The basic YOLOv5 architecture can be divided into four major components:

Input → Backbone → Neck → Detection Head

Input

The input receives and preprocesses steel surface images.

Backbone

The backbone extracts important visual features from the image. YOLOv5 uses a CSP-based structure related to residual-network ideas.

Neck

The neck combines feature maps from different scales. This is important because steel defects may be small, medium, or large.

Detection Head

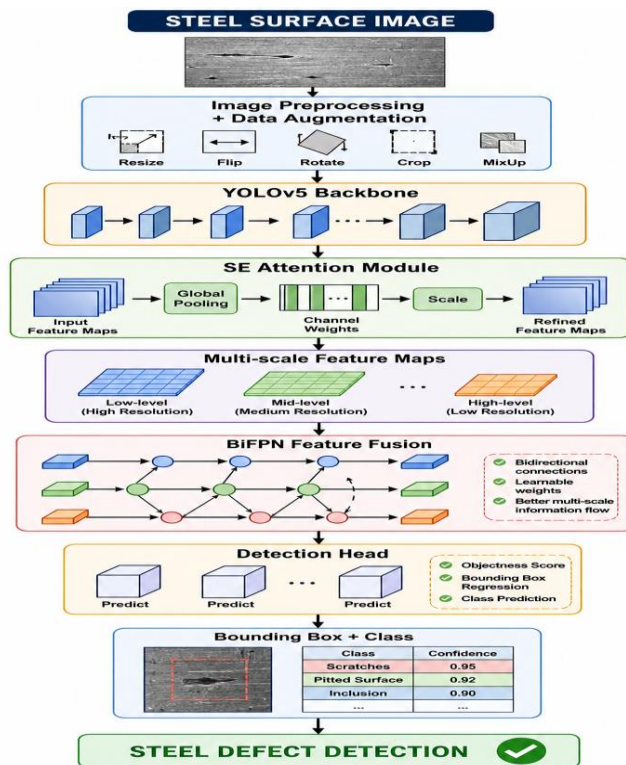
The detection head predicts:

- Defect class
- Bounding-box location
- Confidence score

The original YOLOv5 architecture uses PANet for feature aggregation.

3.4 Proposed YOLOv5-KBS Architecture

The proposed model modifies the YOLOv5l architecture.



The architecture is based on three major improvements.

A. K-means Anchor Optimization

Default YOLO anchor boxes are not necessarily ideal for steel defects because the shapes and sizes of defects are different from general-purpose objects.

The authors therefore apply **K-means clustering** to obtain nine dataset-specific anchor boxes.

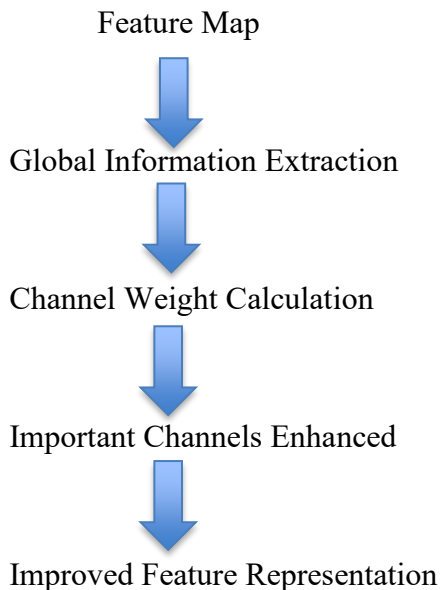
The clustering uses Intersection over Union (IoU) as the distance-related measure. This makes the anchor boxes better suited to the defect distribution.

B. SE Attention Mechanism

The **Squeeze-and-Excitation module** is introduced into the backbone.

Its purpose is to assign greater importance to useful feature channels and reduce the influence of less useful features.

Conceptually:



This is particularly useful when defects occupy only a small portion of an image or when the background contains confusing patterns.

The paper's Grad-CAM analysis indicates that adding SE attention helped the model concentrate more effectively on relevant defect regions.

C. BiFPN

The original PANet neck is replaced with a **Bidirectional Feature Pyramid Network**.

BiFPN performs multi-scale feature fusion using learnable weights. Instead of treating every feature input equally, the network learns which feature information is more important.

This is valuable for steel inspection because the model must detect both:

- Small defects
- Large defects

The weighted feature fusion can therefore improve representation across multiple scales.

3.5 Training and Evaluation

The experiments were implemented using **PyTorch** and conducted using an NVIDIA A16 GPU. The researchers applied augmentation methods to reduce overfitting.

The main evaluation measures were:

Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Precision indicates how many predicted defects are actually correct.

Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Recall measures how many of the actual defects were successfully detected.

mAP

Mean Average Precision is used to evaluate object-detection accuracy across the defect categories.

FPS

Frames Per Second measures detection speed.

For an industrial system, both accuracy and speed are important because a highly accurate model that operates too slowly may not be practical on a production line.

Q4. Results, Comparison and Critical Analysis

4.1 Comparison of Baseline Models

The authors initially compared YOLOv5, YOLOX and YOLOv7.

Model	Average mAP	FPS	General Observation
YOLOv5	75.7%	71 FPS	Best balance among tested baselines

Model	Average mAP	FPS	General Observation
YOLOX	Lower than YOLOv5	Lower than YOLOv5	Accuracy/speed trade-off
YOLOv7	Lower than YOLOv5	55 FPS	Good performance but slower

The article identifies YOLOv5 as the most suitable baseline because it provides a favourable combination of detection accuracy and processing speed.

4.2 Improvement Achieved by YOLOv5-KBS

The proposed YOLOv5-KBS improves the baseline by combining the three main modifications.

Component	Purpose
K-means anchors	Better localization of defect shapes and sizes
SE attention	Focus on useful feature channels
BiFPN	Better multi-scale feature fusion
Data augmentation	Improve generalization
YOLOv5 backbone/head	Fast object detection

The final model achieved a **4.2 percentage-point improvement in mAP** over the YOLOv5l baseline and approximately **70 FPS** detection speed.

4.3 Defect-Level Improvements

The paper reports significant improvements for several defect categories.

Defect Type	Reported Improvement
Scratches	+10.8%
Pitted surface	+9.6%
Crazing	+4.3%

Defect Type	Reported Improvement
Patches	+2.7%
Inclusion	Approximately unchanged
Rolled-in scale	Approximately unchanged

This indicates that the proposed modifications are not equally beneficial for every defect class. The largest improvements occur for scratches and pitted-surface defects.

4.4 Strengths

1. Real-time capability

Approximately 70 FPS is a strong result for industrial inspection because production systems require rapid processing.

2. Multi-scale detection

BiFPN is appropriate because steel defects have different sizes.

3. Attention mechanism

SE helps the network emphasize important channels and reduce the influence of irrelevant background information.

4. Dataset-specific anchor boxes

Using K-means to generate anchors makes the detector more appropriate for the geometry of steel defects.

5. Explainability analysis

The authors use Grad-CAM to examine how the attention mechanism changes the model's focus. This provides some visual evidence supporting the proposed modification.

4.5 Limitations

Despite its positive results, the article has some limitations.

1. Limited dataset diversity

The experiment uses 1,800 images from the NEU-DET dataset. Although the dataset is useful for benchmarking, industrial environments can contain considerably more variation in lighting, camera position, steel type and surface condition.

2. Small-defect problem

The authors themselves identify difficulties with very small defects, particularly inclusions.

3. Overlapping defects

The model may have difficulties when multiple defects overlap or appear close together. The paper specifically identifies overlapping flaws as an area requiring further investigation.

4. Computational requirements

Although the model operates at approximately 70 FPS, it was tested using an NVIDIA A16 GPU. A production environment may require deployment on an edge device with substantially lower computational resources.

5. Generalization

A model trained and tested on one benchmark dataset may not automatically perform equally well on images from another steel factory.

4.6 Critical Evaluation

The article provides a logical methodology because the authors do not simply replace the entire network. Instead, they identify specific weaknesses in the baseline and address them using targeted modifications.

The combination of attention, feature fusion and anchor optimization is technically reasonable. The ablation experiments also strengthen the argument that the individual components contribute to performance improvement.

However, the reported improvement should not be interpreted as proof that the model will outperform all possible industrial inspection systems. The experiments are primarily based on one benchmark dataset. A stronger validation would include multiple industrial datasets and images collected from different production lines.

Another important consideration is the balance between accuracy and computational complexity. Increasing model complexity may improve accuracy but can make deployment more difficult. Therefore, future studies should report parameters, FLOPs, memory usage and inference time on edge hardware in addition to FPS on a high-performance GPU.

4.7 Future Research Directions

At least two improvements can be proposed:

Improvement 1: Lightweight Edge Deployment

A lightweight backbone such as MobileNet or a compressed YOLO architecture could be investigated. Model pruning, quantization and knowledge distillation could also reduce memory and computational requirements.

This would make the system more suitable for deployment near production-line cameras.

Improvement 2: Better Small-Defect Detection

Small defects could be addressed using higher-resolution feature maps, improved feature pyramids, super-resolution preprocessing or specialized small-object detection modules.

Improvement 3: Larger Industrial Dataset

Future research should collect images from multiple factories under different lighting, camera and production conditions.

Improvement 4: Real-Time Video Tracking

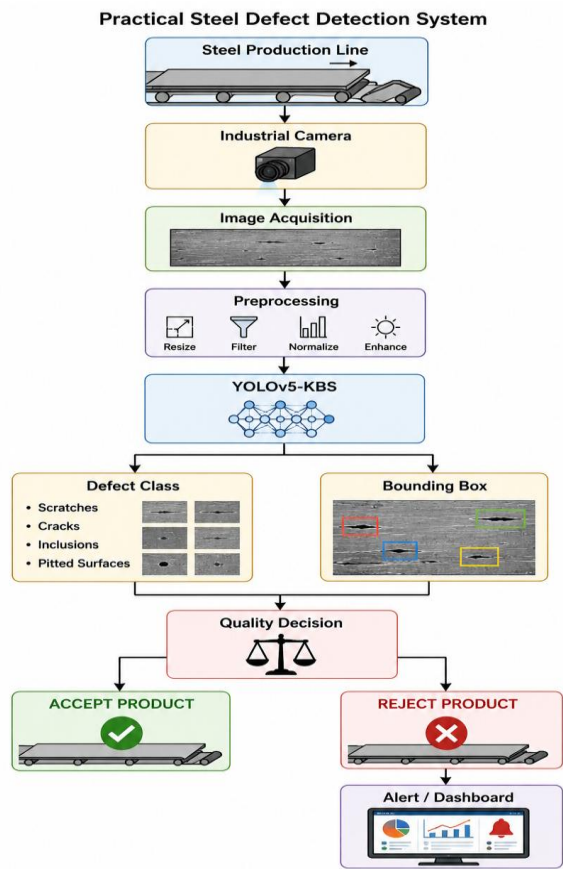
Instead of processing individual frames independently, object tracking could be added. This could provide more stable detection results and reduce repeated alarms for the same defect.

Q5. Application and Personal Evaluation

5.1 Practical Industry Application

The reviewed method can be applied to an automated **steel manufacturing quality inspection system**.

A camera can continuously capture images of steel sheets moving along a production line. The YOLOv5-KBS model can analyse each frame and identify defects.



5.2 Proposed Implementation

For a real industrial implementation, I would propose the following components:

Component	Proposed Choice
Camera	Industrial high-speed camera
Processing	GPU/edge AI device
Detection model	YOLOv5-KBS or lightweight variant
Input	Continuous steel-strip video
Output	Defect class + bounding box + confidence
Decision	Pass/Fail classification
Storage	Defect images and inspection logs

Component	Proposed Choice
Monitoring	Web-based quality dashboard

The system could automatically trigger an alarm when the confidence score of a detected defect exceeds a predefined threshold.

For example:

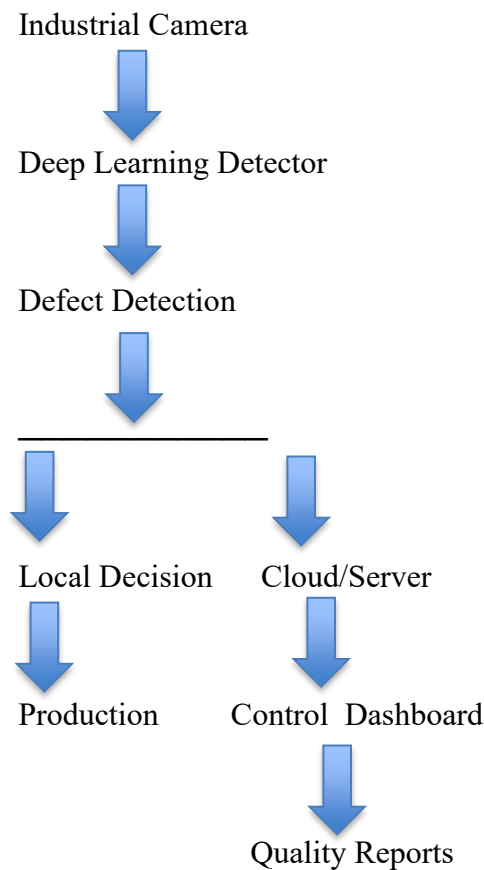
Camera → Detection → Defect Localization → Classification → Decision → Alert

This would reduce dependence on manual inspection and provide continuous monitoring.

5.3 Suggested Extension

For a modern industrial implementation, I would extend the article's approach by combining **YOLOv5-KBS with IoT monitoring**.

The architecture could be:



The system could store:

- Number of defects

- Defect type
- Defect location
- Detection confidence
- Production batch
- Time of detection
- Number of rejected products

This information could then be used for production analysis and predictive quality control.

5.4 Personal Evaluation

In my evaluation, this article is a useful contribution to practical deep learning because it addresses a genuine industrial problem rather than focusing only on theoretical model development.

The strongest aspect of the research is its focus on the **accuracy-speed trade-off**. Industrial inspection systems cannot depend only on high accuracy; they must also process products rapidly. The reported approximately 70 FPS performance makes the proposed method attractive for real-time inspection.

The use of SE attention is also meaningful because steel surfaces can contain complicated backgrounds. BiFPN is appropriate for multi-scale defect detection, while K-means anchor optimization adapts the detector to the specific geometry of steel defects.

However, I believe the method requires additional validation before large-scale industrial deployment. The dataset is relatively small compared with the diversity of real manufacturing environments. More testing on different factories, cameras, steel grades and lighting conditions would improve confidence in the model's generalization.

The model could also be made more practical through pruning, quantization and lightweight architectures. This would allow deployment on edge devices directly beside production-line cameras.

Overall, the article demonstrates how an established deep learning architecture can be systematically improved for a specific industrial application. Its combination of **YOLOv5, SE attention, BiFPN and K-means anchor optimization** provides a practical solution for automated steel surface inspection.

Conclusion

The reviewed research paper presents an improved deep learning approach for steel surface defect detection using the YOLOv5-KBS architecture. The study demonstrates that modifying an established object detection model can improve its performance for a specialized industrial task.

The proposed architecture addresses three important challenges: irrelevant background information, defects with different sizes and inappropriate default anchor boxes. SE attention improves feature selection, BiFPN improves multi-scale feature fusion, and K-means clustering generates more suitable anchor boxes.

Experimental results demonstrate a 4.2 percentage-point mAP improvement over the YOLOv5 baseline while maintaining approximately 70 FPS detection speed. The improvements are particularly noticeable for scratches and pitted-surface defects.

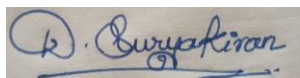
The research is therefore relevant to Industry 4.0 and automated manufacturing quality control. Future work should focus on larger and more diverse industrial datasets, small-defect detection, overlapping defects, lightweight edge deployment, and real-world production-line validation.

References

1. Gao, Y., Lv, G., Xiao, D., Han, X., Sun, T., & Li, Z. (2024). **Research on steel surface defect classification method based on deep learning.** *Scientific Reports*, 14, 8254. <https://doi.org/10.1038/s41598-024-58643-1>.
2. Hu, J., Shen, L., & Sun, G. (2018). **Squeeze-and-excitation networks.** *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 7132–7141.
3. Tan, M., Pang, R., & Le, Q. V. (2020). **EfficientDet: Scalable and efficient object detection.** *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 10781–10790.
4. Lin, T. Y., et al. (2017). **Feature pyramid networks for object detection.** *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2117–2125.
5. Liu, S., et al. (2018). **Path aggregation network for instance segmentation.** *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 8759–8768.

Code of Conduct:

I D. Suryakiran/192325057 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Understanding & Summary	20%	Comprehensive and accurate understanding of the article, problem, objectives, and contributions.	Good understanding with minor omissions or inaccuracies.	Basic understanding; important details are missing.	Poor understanding or significant misinterpretation.
Critical Review of Methodology	25%	Thoroughly evaluates methodology, architecture, datasets, training, and evaluation choices.	Appropriate analysis with minor gaps.	Limited analysis with several unexplained aspects.	Little or no meaningful analysis of methodology.
Analysis of Results	20%	Accurately interprets results, comparisons, metrics, strengths, and limitations.	Good interpretation with reasonable comparison.	Basic interpretation with limited comparison.	Incorrect, unsupported, or missing interpretation.
Application & Proposed Improvements	20%	Practical, innovative application with well-justified improvements and future directions.	Relevant application and feasible improvements.	Basic application with limited justification.	Unclear, impractical, or unsupported proposal.
Presentation & Referencing	15%	Well-organized, professional presentation with clear diagrams.	Clear presentation with minor formatting or referencing issues.	Adequate presentation with inconsistent formatting/referencing.	Poor organization, unclear presentation, or inadequate referencing.

Suggested Article Review Structure

1. Article Citation
2. Abstract / Executive Summary
3. Research Problem and Motivation
4. Methodology and Deep Learning Architecture
5. Dataset and Experimental Setup
6. Results and Comparative Analysis
7. Strengths and Limitations
8. Proposed Improvements / Future Scope
9. Practical Industry Application
10. Conclusion and References